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Fuel cell ship

Abstract

A fuel cell ship includes a cooling system that cools a fuel cell. The cooling system includes a cooling medium tank that accommodates a cooling medium, a cooling medium circulation pipe that circulates the cooling medium between the fuel cell and the cooling medium tank, a cooling tank internal gas detector installed in the cooling medium tank, a cooling tank internal gas discharge pipe connected to the cooling medium tank, and a cooling tank internal gas discharge valve installed in the cooling tank internal gas discharge pipe. The fuel cell ship includes a control unit that controls opening and closing of the cooling tank internal gas discharge valve. The control unit opens the cooling tank internal gas discharge valve when the cooling tank internal gas detector detects that the concentration of the fuel gas in the cooling medium tank is equal to or greater than a specified value determined in advance.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0178920	12/2004	Wilson	244/171.5	B64G 1/409

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201409006	12/2009	CN	N/A
2579374	12/2012	EP	N/A
2004319206	12/2003	JP	N/A
2018-092815	12/2017	JP	N/A

OTHER PUBLICATIONS

European Search Report dated Jun. 6, 2023 issued in EP Application 22175803.0. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is claims priority under 35 U.S.C. § 119 to JP Application No. 2021-092709 filed Jun. 2, 2021, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a fuel cell ship.

BACKGROUND ART

(3) In the related art, a fuel cell ship in which a fuel gas (for example, hydrogen gas) is supplied from a fuel tank to a fuel cell and a propulsion device is driven by electric power generated by the

fuel cell has been proposed (see Patent Document 1, for example).

PRIOR ART DOCUMENT

Patent Document

(4) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2018-92815

SUMMARY OF INVENTION

Technical Problem

(5) Fuel cells generate heat while generating electric power. To maintain an appropriate power generation efficiency in a fuel cell, it is desirable to cool the fuel cell by supplying, for example, a cooling medium (for example, cooling water). However, in such a configuration, when a fuel gas leak occurs in the fuel cell for some reason, the leaking fuel gas may enter a cooling medium tank through pipes for circulating the cooling medium. Considering that the fuel gas is a combustible gas and there is thus a risk of explosion, it is not desirable that the leaking fuel gas is retained in the cooling medium tank.

(6) The present invention has been contrived to solve the above-described problems, and an object thereof is to provide a fuel cell ship capable of preventing leaking fuel gas from entering and staying in a cooling medium tank, when a fuel gas leak occurs in a fuel cell.

Solution to Problem

(7) A fuel cell ship according to one aspect of the present invention includes a fuel cell that generates electric power by an electrochemical reaction of a fuel and a propulsion device that generates a propulsive force on a hull by the electric power supplied from the fuel cell, the fuel cell ship further includes a cooling system that cools the fuel cell, and the cooling system includes a cooling medium tank that accommodates a cooling medium, a cooling medium circulation pipe that circulates the cooling medium between the fuel cell and the cooling medium tank, a cooling tank internal gas detector installed in the cooling medium tank, a cooling tank internal gas discharge pipe connected to the cooling medium tank, and a cooling tank internal gas discharge valve installed in the cooling tank internal gas discharge pipe, the fuel cell ship further including a control unit that controls opening and closing of the cooling tank internal gas discharge valve, and the control unit opening the cooling tank internal gas discharge valve when the cooling tank internal gas detector detects that a concentration of a fuel gas being a gaseous state of the fuel is equal to or greater than a specified value determined in advance in the cooling medium tank.

Advantageous Effects of Invention

(8) According to the above-described configuration, when a fuel gas leak occurs in a fuel cell, it is possible to prevent leaking fuel gas from entering and staying in a cooling medium tank.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an explanatory diagram illustrating a schematic configuration of a fuel cell ship according to an embodiment of the present invention.

(2) FIG. 2 is an explanatory diagram schematically illustrating an internal structure of the fuel cell ship.

(3) FIG. 3 is an explanatory diagram schematically illustrating a schematic configuration of a fuel cell included in the fuel cell ship.

(4) FIG. 4 is a flowchart illustrating a processing flow according to control for opening and closing a cooling tank internal gas discharge valve included in the fuel cell ship.

(5) FIG. 5 is an explanatory diagram schematically illustrating a configuration of main parts of the fuel cell ship.

(6) FIG. 6 is an explanatory diagram schematically illustrating another configuration of main parts of the fuel cell ship.

DESCRIPTION OF EMBODIMENTS

(7) An embodiment of the present invention will be described below with reference to the drawings. Herein, directions are defined as follows. First, a direction directed from a stern to a bow of a fuel cell ship is “front”, and a direction directed from the bow to the stern is “rear”. A horizontal direction perpendicular to a front-rear direction is defined as a left-right direction. At this time, when the fuel cell ship is moving forward, the left side is defined as “left” and the right side is defined as “right”, when viewed from an operator. An upstream side in a gravity direction perpendicular to the front-rear direction and the left-right direction, is referred to as “up”, and a downstream side is referred to as “down”.

(8) [1. Schematic Configuration of Fuel Cell Ship]

(9) First, a fuel cell ship SH according to the present embodiment will be described with reference to FIG. 1. FIG. 1 is an explanatory diagram illustrating a schematic configuration of the fuel cell ship SH. The fuel cell ship SH includes a hull 1 and a cabin 2. The cabin 2 is arranged on an upper surface of the hull 1.

(10) The fuel cell ship SH further includes a fuel cell system 3, a fuel gas storage unit 4, a storage battery system 5, a propulsion device 6, a cooling system 7, a plurality of pieces of peripheral equipment 11, and a control device 12. In FIG. 1, a control signal or a high-voltage power supply line is indicated by a solid line, and a control signal or a low-voltage power supply line is indicated by a dashed line.

(11) The fuel cell system 3 functions as a main power supply. The fuel cell system 3 consumes a fuel gas to generate electric power (specifically, DC electric power). The fuel gas is an example of a fuel, for example, a combustible gas. Typically, the fuel gas is hydrogen gas. The fuel cell system 3 supplies generated electric power to the propulsion device 6 and the peripheral equipment 11. The fuel cell system 3 can also supply electric power to the storage battery system 5 to charge the storage battery system 5. Details of the fuel cell system 3 will be described later.

(12) The fuel gas storage unit 4 stores the fuel gas to be supplied to the fuel cell system 3. Specifically, the fuel gas storage unit 4 includes a fuel tank 41 (see FIG. 2) that accommodates fuel gas as fuel. The fuel gas is supplied from the fuel tank 41 to the fuel cell system 3 via a fuel gas supply pipe 32 described later (see FIG. 2).

(13) The storage battery system 5 includes a storage battery. The storage battery is, for example, a lithium secondary battery, but may also be a nickel-cadmium storage battery, a nickel-hydrogen storage battery, or the like. The storage battery system 5 functions as an auxiliary power source that supplies stored electric power (specifically, DC electric power) to the propulsion device 6 and the peripheral equipment 11. When the storage battery system 5 functions as the auxiliary power source as described above, it is possible to compensate for a shortage of electric power supplied from the fuel cell system 3 to the propulsion device 6 or the like. The storage battery system 5 may supply electric power to the control device 12.

(14) The propulsion device 6 is driven by electric power supplied from a fuel cell 31 (described later) (see FIG. 2) of the fuel cell system 3, and generates a propulsive force on the hull 1. That is, the fuel cell ship SH includes the propulsion device 6 that generates the propulsive force on the hull 1 by the electric power supplied from the fuel cell 31.

(15) It is noted that the propulsion device 6 may be driven only by the electric power supplied from the storage battery included in the storage battery system 5, or may be driven by the electric power supplied from both the fuel cell 31 and the storage battery. That is, the propulsion device 6 may be driven by the electric power supplied from at least one of the fuel cell 31 and the storage battery to generate the propulsive force on the hull 1.

(16) The propulsion device 6 includes an electric power conversion device 6a, a propulsion motor 6b, and a propeller 6c. The electric power conversion device 6a converts the electric power supplied from the fuel cell system 3 into electric power according to the specifications of the propulsion motor 6b. For example, the electric power conversion device 6a converts DC electric

power into AC electric power. In this case, the electric power conversion device **6a** includes an inverter, for example. The propulsion motor **6b** is driven by electric power (for example, AC electric power) supplied from the electric power conversion device **6a**. When the propulsion motor **6b** is driven, a rotational force of the propulsion motor **6b** is transmitted to the propeller **6c**. As a result, the propeller **6c** rotates, and a propulsive force is generated on the hull **1**. A configuration is also possible in which a marine gear is provided between the propulsion motor **6b** and the propeller **6c**.

(17) Examples of the peripheral equipment **11** include a compressor, a solenoid valve, and a pump. Examples of the peripheral equipment **11** also include electrical equipment such as lighting equipment and air conditioning equipment, but the types of the peripheral equipment **11** are not particularly limited.

(18) The control device **12** controls the fuel cell system **3**, the fuel gas storage unit **4**, the storage battery system **5**, the propulsion device **6**, the cooling system **7**, and the plurality of pieces of peripheral equipment **11**. The control device **12** is configured by one or two or more computers, for example. The computer is, for example, a programmable logic controller (PLC), but may also be an electronic control unit (ECU). The control device **12** is supplied with electric power from a battery (for example, a lead battery) (not illustrated) or the storage battery of the storage battery system **5**.

(19) The control device **12** includes a control unit **12a** and a storage unit **12b**. The control unit **12a** includes a processor such as a central processing unit (CPU). The storage unit **12b** includes a storage device and stores data and computer programs. Specifically, the storage unit **12b** includes a main storage device such as a semiconductor memory and an auxiliary storage device such as a semiconductor memory, a solid state drive, and/or a hard disk drive. The storage unit **12b** may also include removable media. The storage unit **12b** corresponds to an example of a non-transitory computer-readable storage medium.

(20) The processor of the control unit **12a** executes a computer program stored in the storage device of the storage unit **12b**, to control the fuel cell system **3**, the fuel gas storage unit **4**, the storage battery system **5**, the propulsion device **6**, the cooling system **7**, and the plurality of pieces of peripheral equipment **11**.

(21) The cooling system **7** has a function of cooling the fuel cell system **3**, and in particular, the fuel cell **31**. That is, the fuel cell ship SH according to the present embodiment includes the cooling system **7** for cooling the fuel cell **31**. Details of the cooling system **7** will be described later.

(22) [2. Details of Fuel Cell System]

(23) Next, the details of the fuel cell system **3** will be described. FIG. 2 is an explanatory diagram schematically illustrating an internal structure of the fuel cell ship SH.

(24) The fuel cell ship SH includes an engine room **13** and a fuel room **14**. The engine room **13** and the fuel room **14** are arranged below a deck **1a** of the hull **1**. In other words, the engine room **13** and the fuel room **14** are arranged between the deck **1a** and a bottom plate **1b** of the hull **1**. The bottom plate **1b** is located between the deck **1a** and a ship bottom unit **1c** (see FIG. 1).

(25) The engine room **13** is located on the bow side with respect to the fuel room **14**. The engine room **13** and the fuel room **14** are separated by a partition wall (not illustrated). The partition wall is formed of fiber reinforced plastics (FRP), for example, but may also be formed of an iron plate. The fuel tank **41** of the fuel gas storage unit **4** described above is located in the fuel room **14**. As described above, the fuel cell ship SH includes the fuel tank **41** that accommodates fuel.

(26) The fuel cell system **3** of the fuel cell ship SH is located in the engine room **13**. The fuel cell system **3** includes the fuel cell **31**, the fuel gas supply pipe **32**, and a fuel cell side shutoff valve **33**. The fuel cell side shutoff valve **33** is an example of the peripheral equipment **11** (see FIG. 1).

(27) The fuel cell **31** generates electric power (specifically, DC electric power) by an electrochemical reaction between the fuel gas being an example of fuel and an oxidant gas. Typically, the oxidant gas is air and the oxidant is oxygen. That is, the fuel cell ship SH includes the fuel cell **31** that generates electric power by an electrochemical reaction of fuel.

(28) FIG. 3 is an explanatory diagram schematically illustrating a schematic configuration of the fuel cell **31**. The fuel cell **31** is, for example, a polymer electrolyte fuel cell (PEFC), and is formed by a fuel cell stack obtained by stacking a plurality of cells **310**. In FIG. 3, for simplification, the fuel cell **31** is illustrated as being composed of two cells **310**. Each cell **310** of the fuel cell **31** includes a solid polymer electrolyte membrane **311**, an anode electrode **312**, a cathode electrode **313**, and a pair of separators **314a** and **314b**.

(29) The solid polymer electrolyte membrane **311** is sandwiched between the anode electrode **312** and the cathode electrode **313**. The anode electrode **312** is a negative electrode (fuel electrode). The anode electrode **312** includes an anode catalyst layer and a gas diffusion layer. The cathode electrode **313** is a positive electrode (air electrode). The cathode electrode **313** includes a cathode catalyst layer and a gas diffusion layer. The anode electrode **312**, the solid polymer electrolyte membrane **311**, and the cathode electrode **313** form a membrane-electrode assembly (MEA). The pair of separators **314a** and **314b** sandwich the membrane-electrode assembly.

(30) Each of the separators **314a** and **314b** is made of stainless steel (SUS), for example, and includes ribs having a concave-convex shape and forming a plurality of grooves on both sides. Each groove located on one side (the side facing the anode electrode **312**) of the separator **314a** forms a flow channel CH1 for the fuel gas. Each groove located on one side (the side facing the cathode electrode **313**) of the separator **314b** forms a flow channel CH2 for the oxidant gas. Each groove on the other side of the separator **314b** of any one of the cells **310** (for example, referred to as a cell **310B**) and each groove on the other side of the separator **314a** of a cell **310** adjacent to the one cell **310** (for example, referred to as a cell **310A**) form a flow channel CH3 for a cooling medium described later. Various types of detectors (pressure sensor, temperature sensor, etc.) are provided in the flow channel CH1 and the flow channel CH3, which will be described later.

(31) In FIG. 3, the flow channel CH3 of the cooling medium is provided between adjacent ones of the cells **310**, but it is not necessarily required that the flow channel CH3 is sandwiched between the cells **310**. A configuration may be provided in which the flow channel CH3 of the cooling medium is sandwiched for each stack including several cells, as long as each cell **310** can be cooled to an appropriate temperature.

(32) In the above-described configuration of the fuel cell **31**, hydrogen contained in the fuel gas flowing through the flow channel CH1 is decomposed into hydrogen ions and electrons by the catalyst on the side of the anode electrode **312** in the cell **310B**, for example. Hydrogen ions pass through the solid polymer electrolyte membrane **311** and move to the side of the cathode electrode **313**. The concave-convex ribs of the separator **314a** of the cell **310B** contact the anode electrode **312** by sandwiching the flow channel CH1, and thus, electrons generated at the anode electrode **312** move to the separator **314a**. The separator **314a** of the cell **310B** and the separator **314b** sandwiching the flow channel CH3 of the cooling medium with the separator **314a** contact each other by the ribs, and thus, the above-mentioned electrons move from the separator **314a** to the separator **314b**. After moving to the separator **314b**, the electrons pass through an external circuit **315** and move to the separator **314a** at the opposite end in a stacking direction. Thus, an electric current is generated (electricity is generated).

(33) The separator **314a** and the separator **314b** of the cell **310A** sandwiching the flow channel CH3 of the cooling medium with the separator **314a** contact each other by the ribs, and thus, the above-mentioned electrons move from the separator **314a** to the separator **314b** of the cell **310A**. The concave-convex ribs of the separator **314b** contact the cathode electrode **313** by sandwiching the flow channel CH2 of the oxidant gas, and thus, after moving to the separator **314b**, the electrons move to the cathode electrode **313**. On the side of the cathode electrode **313**, oxygen contained in the oxidant gas flowing through the flow channel CH2 combines with the above-mentioned electrons and hydrogen ions passing through the solid polymer electrolyte membrane to generate water. The generated water is discharged outside the ship via a discharge pipe **31a** (see FIG. 2).

(34) The fuel cell **31** supplies generated electric power to the propulsion device **6** and the peripheral equipment **11** illustrated in FIG. **1**. The fuel cell **31** may indirectly supply the generated electric power to the propulsion device **6** and the peripheral equipment **11** via a circuit such as a DC/DC converter or the like.

(35) The fuel gas supply pipe **32** illustrated in FIG. **2** is a fuel supply pipe for supplying, to the anode electrode **312** (see FIG. **3**) of the fuel cell **31**, the fuel gas accommodated in the fuel tank **41** of the fuel gas storage unit **4**. That is, the fuel cell ship SH includes the fuel gas supply pipe **32** as a fuel supply pipe through which fuel is supplied from the fuel tank **41** to the fuel cell **31**.

(36) The fuel cell side shutoff valve **33** is a shutoff valve that opens or closes the flow channel of the fuel gas supply pipe **32**. The opening and closing of the fuel cell side shutoff valve **33** is controlled by the control unit **12a** (see FIG. **1**). More specifically, the fuel cell side shutoff valve **33** switches between supplying and stopping the supply of fuel gas from the fuel tank **41** to the fuel cell **31**, based on the control by the control unit **12a**. Although only one fuel cell side shutoff valve **33** is provided in the fuel gas supply pipe **32** in a fuel cell compartment **30** described later, two or more of the fuel cell side shutoff valves **33** may be provided.

(37) The fuel cell ship SH further includes the fuel cell compartment **30**. The fuel cell compartment **30** is a housing body that houses the fuel cell **31**, and is arranged in the engine room **13**.

(38) The fuel cell compartment **30** has a hollow shape. For example, the fuel cell compartment **30** has a hollow and substantially rectangular parallelepiped shape. In this case, outer walls constituting the fuel cell compartment **30** include, for example, a top wall **30a**, a bottom wall **30b**, a front wall (not illustrated), a rear wall (not illustrated), a side wall **30c**, and a side wall **30d**. However, the top surface, bottom surface, front surface, rear surface, and side surfaces of the fuel cell compartment **30** can be freely determined. The shape of the fuel cell compartment **30** is not particularly limited, as long as the fuel cell compartment **30** has a space that can house the fuel cell **31**. The fuel cell compartment **30** can also be considered as a container, a chamber, or a box that houses the fuel cell **31**. The material of the outer wall of the fuel cell compartment **30** is FRP, for example, but may be an iron plate.

(39) A cell compartment air supply port **30e** with an opening is provided in the side wall **30d** of the fuel cell compartment **30**. The cell compartment air supply port **30e** is connected to a cell compartment air supply pipe **35**. The cell compartment air supply pipe **35** extends from the cell compartment air supply port **30e** to the deck **1a** and is exposed from an upper surface of the deck **1a**. The cell compartment air supply port **30e** may be provided in an outer wall other than the side wall **30d** in the fuel cell compartment **30**.

(40) On the other hand, a cell compartment exhaust port **30f** with an opening is provided in the side wall **30c** of the fuel cell compartment **30**. The cell compartment exhaust port **30f** is connected to a communication unit **36**. The communication unit **36** communicates with a duct compartment (not illustrated) forming an exhaust passage. The duct compartment communicates with a vent pipe (not illustrated) that communicates with the outside of the ship. Thus, air entering from the cell compartment air supply pipe **35** via the cell compartment air supply port **30e** into the fuel cell compartment **30** is discharged outside the ship via the cell compartment exhaust port **30f**, the communication unit **36**, the duct compartment, and the vent pipe. As a result, the inside of the fuel cell compartment **30** is ventilated.

(41) The fuel cell compartment **30** has an interior that is a closed space, except for the cell compartment air supply port **30e** and the cell compartment exhaust port **30f**.

(42) A part of the fuel gas supply pipe **32** described above and the fuel cell side shutoff valve **33** are housed in the fuel cell compartment **30**. A cell compartment internal gas detector **34a** and a cell compartment internal fire detector **34b** are further housed in the fuel cell compartment **30**.

(43) The cell compartment internal gas detector **34a** is a fuel gas detector arranged inside the fuel cell compartment **30**. For example, if the fuel gas is hydrogen gas, the cell compartment internal gas detector **34a** is composed of a hydrogen gas detection sensor.

(44) The cell compartment internal gas detector **34a** is arranged on an inner surface of the top wall **30a** located at an upper part of the fuel cell compartment **30**. Hydrogen gas as a fuel gas is lighter than air and rises. Therefore, by arranging the cell compartment internal gas detector **34a** on the top wall **30a** of the fuel cell compartment **30**, leaking fuel gas can be reliably detected by the cell compartment internal gas detector **34a**, even if the fuel gas leaks in the fuel cell compartment **30**. The installation position of the cell compartment internal gas detector **34a** may be configured to be located furthestmost downstream of the flow channel through which the fuel gas flows when fuel gas leaks in the fuel cell compartment **30**.

(45) If the cell compartment internal gas detector **34a** detects fuel gas in the fuel cell compartment **30**, a detection signal is transmitted from the cell compartment internal gas detector **34a** to the control unit **12a**. Accordingly, the control unit **12a** can control the fuel cell side shutoff valve **33** provided in the fuel gas supply pipe **32** to stop the supply of fuel gas from the fuel tank **41** to the fuel cell **31**.

(46) The cell compartment internal fire detector **34b** is a fire detector arranged inside the fuel cell compartment **30**. The cell compartment internal fire detector **34b** includes, for example, one or more sensors among a smoke sensor that detects smoke, a heat sensor that detects heat, and a flame sensor that detects a flame. The cell compartment internal fire detector **34b** may be configured by a thermocouple type fire detector.

(47) The cell compartment internal fire detector **34b** is arranged on an inner surface of the top wall **30a** located at an upper part of the fuel cell compartment **30**. In the unlikely event that a fire occurs inside the fuel cell compartment **30**, the cell compartment internal fire detector **34b** detects the fire and outputs a detection signal indicating that a fire has occurred to the control unit **12a**. In this case, the control unit **12a** can control the fuel cell side shutoff valve **33** to stop the supply of fuel gas from the fuel tank **41** to the fuel cell **31**. Thus, in the fuel cell compartment **30**, the risk of explosion due to ignition of the fuel gas can be reduced as much as possible.

(48) The above-mentioned fuel cell system **3** will be further described. The fuel cell system **3** includes an oxidant gas flow rate adjustment unit **321**, an off-gas circulation unit **322**, a gas-liquid separation unit **323**, and a discharge unit **324**. The oxidant gas flow rate adjustment unit **321**, the off-gas circulation unit **322**, and the discharge unit **324** are examples of the peripheral equipment **11**. The control unit **12a** controls the oxidant gas flow rate adjustment unit **321**, the off-gas circulation unit **322**, and the discharge unit **324**. The fuel cell system **3** further includes the discharge pipe **31a**, an oxidant gas pipe **31b**, an off-gas circulation pipe **31c**, and a connection pipe **31d**. A manifold for circulating the fuel gas, the oxidant gas, and the cooling medium described later is formed inside the fuel cell **31**.

(49) The oxidant gas flow rate adjustment unit **321** supplies the oxidant gas to the cathode electrode **313** (see FIG. 3) of the fuel cell **31**. Specifically, the oxidant gas flow rate adjustment unit **321** adjusts the flow rate of the oxidant gas supplied to the fuel cell **31**. Typically, the oxidant gas flow rate adjustment unit **321** is an air compressor that compresses the oxidant gas.

(50) The oxidant gas pipe **31b** guides the oxidant gas supplied from the oxidant gas flow rate adjustment unit **321** to the cathode electrode **313** of the fuel cell **31**.

(51) The above-mentioned discharge pipe **31a** is connected to a discharge manifold on the side of the cathode electrode **313** provided inside the fuel cell **31**. The discharge pipe **31a** guides oxidant off-gas and water discharged from the fuel cell **31** to the atmosphere. The oxidant off-gas refers to an exhaust gas from the cathode electrode **313**. That is, the oxidant off-gas is a cathode off-gas.

(52) The gas-liquid separation unit **323** separates water contained in the fuel off-gas discharged from the fuel cell **31**, and discharges the water to the connection pipe **31d**. In addition, the gas-liquid separation unit **323** discharges, to the off-gas circulation pipe **31c**, excess fuel gas, which is the fuel off-gas after the water is separated. Typically, the gas-liquid separation unit **323** is a gas-liquid separator. The fuel off-gas refers to an exhaust gas from the anode electrode **312** of the fuel cell **31** (see FIG. 3). That is, the fuel off-gas is an anode off-gas.

(53) The off-gas circulation unit **322** is arranged in the off-gas circulation pipe **31c**. The off-gas circulation unit **322** discharges the excess fuel gas discharged from the gas-liquid separation unit **323** to the fuel gas supply pipe **32**. Subsequently, the fuel gas supply pipe **32** supplies the excess fuel gas to the fuel cell **31**. Typically, the off-gas circulation unit **322** is a pump. For example, the off-gas circulation unit **322** may be an ejector.

(54) The discharge unit **324** is arranged in the connection pipe **31d**. The discharge unit **324** discharges the water separated by the gas-liquid separation unit **323**. In addition, the discharge unit **324** discharges a part of the fuel off-gas discharged from the fuel cell **31**, that is, the remaining gas and water that are not supplied to the off-gas circulation pipe **31c**. Typically, the discharge unit **324** is a purge valve.

(55) The water and fuel off-gas discharged from the discharge unit **324** are discharged to the discharge pipe **31a** via the connection pipe **31d**, and are guided to the atmosphere from the discharge pipe **31a** together with the oxidant off-gas (cathode off-gas) discharged from the fuel cell **31**.

(56) [3. Details of Cooling System]

(57) Next, details of the cooling system **7** will be described, still with reference to FIG. **2**. The cooling system **7** includes a cooling medium tank **71**, a cooling medium circulation pipe **72**, a cooling tank internal gas detector **73**, a cooling tank internal gas discharge pipe **74**, and a cooling tank internal gas discharge valve **75**.

(58) The cooling medium tank **71** is a container that accommodates the cooling medium. The cooling medium is, for example, an antifreeze liquid having low electrical conductivity. The antifreeze liquid is, for example, a liquid obtained by mixing pure water and ethylene glycol in a predetermined ratio.

(59) The cooling medium tank **71** is installed inside the engine room **13** and outside the fuel cell compartment **30**. In other words, when the cooling medium tank **71** is installed in the engine room **13** as described above, the engine room **13** forms a cooling medium installation compartment. Therefore, it can be said that the fuel cell ship SH includes the cooling medium installation compartment (the engine room **13**) in which the cooling medium tank **71** is installed.

(60) The fuel gas supply pipe **32** described above extends from the fuel room **14** to the engine room **13** and is connected to the fuel cell **31** of the fuel cell compartment **30**. That is, the fuel gas supply pipe **32** passes through the cooling medium installation compartment.

(61) The engine room **13**, which is the cooling medium installation compartment, includes a ventilation port **13a** that communicates with the outside of the ship. The ventilation port **13a** is an air supply/exhaust port formed by penetrating the deck **1a** forming the upper wall of the engine room **13**. In the present embodiment, the ventilation port **13a** is provided on each of a starboard side and a port side of the hull **1**. When an air compressor (not illustrated) in the engine room **13** is not driven, for example, air from outside the ship is introduced into the engine room **13** from the ventilation port **13a** on the starboard side. Subsequently, the air inside the engine room **13** is discharged outside the ship from the ventilation port **13a** on the port side. On the other hand, air from outside the ship may be introduced into the engine room **13** from the ventilation port **13a** on the port side and discharged outside the ship from the ventilation port **13a** on the starboard side. In both of the above-mentioned air flow patterns, the inside of the engine room **13** is ventilated. In addition, intake and exhaust of air may be performed only by the ventilation port **13a** on the starboard side, and intake and exhaust of air may be performed only by the ventilation port **13a** on the port side. On the other hand, when the air compressor in the engine room **13** is driven, the air compressor sucks air into the engine room **13** through the ventilation ports **13a** on the starboard side and the port side.

(62) The cooling medium circulation pipe **72** is a pipe for circulating the cooling medium between the fuel cell **31** and the cooling medium tank **71**. A cooling medium circulation unit **76** and a heat exchanger **77** are arranged partway along the cooling medium circulation pipe **72**. The cooling

cooling medium circulation unit **76** is formed by a pump, for example. When the cooling medium circulation unit **76** is driven, the cooling medium flows through the cooling medium circulation pipe **72**, and the cooling medium circulates between the fuel cell **31** and the cooling medium tank **71**. When the cooling medium is supplied to the fuel cell **31**, the fuel cell **31** is cooled. The cooling medium used for cooling the fuel cell **31** passes through the cooling medium circulation pipe **72**, is cooled by heat exchange performed by the heat exchanger **77**, and then, is returned to the cooling medium tank **71**. The heat exchanger **77** may be installed between the cooling medium tank **71** and the cooling medium circulation unit **76** in the cooling medium circulation pipe **72**.

(63) In FIG. **2**, the cooling medium circulation unit **76** is arranged in the fuel cell compartment **30**. However, the cooling medium circulation unit **76** may be arranged in the engine room **13** and outside the fuel cell compartment **30**.

(64) As described above, the cooling system **7** includes the cooling medium tank **71** accommodating the cooling medium, and the cooling medium circulation pipe **72** for circulating the cooling medium between the fuel cell **31** and the cooling medium tank **71**.

(65) The cooling tank internal gas detector **73** is a fuel gas detector that is installed on top inside the cooling medium tank **71** to detect the fuel gas existing inside the cooling medium tank **71**. For example, if the fuel gas is hydrogen gas, the cooling tank internal gas detector **73** is configured by a hydrogen gas detection sensor.

(66) An example of the fuel gas existing in the cooling medium tank **71** may include a fuel gas leaking from the fuel cell **31** and entering the cooling medium tank **71** via the cooling medium circulation pipe **72**. A result of detecting the fuel gas by the cooling tank internal gas detector **73** (for example, information about the concentration of the fuel gas) is transmitted to the control unit **12a**. Thus, the control unit **12a** determines, based on the detection result of the cooling tank internal gas detector **73**, whether there is a fuel gas leak in the fuel cell **31**, and if there is a leak, the control unit **12a** can, for example, perform control to stop electric power generation by the fuel cell **31**.

(67) The cooling tank internal gas discharge pipe **74** is connected to the cooling medium tank **71** and is a pipe for discharging the fuel gas existing inside the cooling medium tank **71** to the outside. The outlet of the cooling tank internal gas discharge pipe **74**, that is, the end of the cooling tank internal gas discharge pipe **74** opposite to the connection side with the cooling medium tank **71** forms a gas discharge port **74a** for discharging the fuel gas.

(68) The gas discharge port **74a** is located above electrical equipment EM in the cooling medium installation compartment (the engine room **13**). The above-mentioned ventilation port **13a** of the engine room **13** is located above the gas discharge port **74a**. The reason for defining such a positional relationship will be described later. The electrical equipment EM is equipment constituting the peripheral equipment **11**. Specific examples of the electrical equipment EM may include an air compressor, a junction box that relays the generated electric power of the fuel cell **31** to the electric power conversion device **6a** (see FIG. **1**), an inverter, a converter, and the like.

(69) The cooling tank internal gas discharge valve **75** is installed in the cooling tank internal gas discharge pipe **74** to open or close the flow channel for the fuel gas flowing through the cooling tank internal gas discharge pipe **74**. The opening and closing of the cooling tank internal gas discharge valve **75** is controlled by the control unit **12a**.

(70) As described above, the cooling system **7** includes the cooling tank internal gas detector **73** installed in the cooling medium tank **71**, the cooling tank internal gas discharge pipe **74** connected to the cooling medium tank **71**, and the cooling tank internal gas discharge valve **75** installed in the cooling tank internal gas discharge pipe **74**. In addition, the fuel cell ship SH includes the control unit **12a** that controls the opening and closing of the cooling tank internal gas discharge valve **75**.

(71) Next, details of the control for opening and closing the cooling tank internal gas discharge valve **75** by the control unit **12a** will be described with reference to FIG. **4**. FIG. **4** is a flowchart illustrating a processing flow according to the control for opening and closing the cooling tank

internal gas discharge valve **75**.

(72) When the cooling tank internal gas detector **73** detects that the concentration of the fuel gas (for example, hydrogen gas) in the cooling medium tank **71** is equal to or greater than a specified value (Yes in **S1**), the control unit **12a** opens the cooling tank internal gas discharge valve **75** (**S2**). In this case, the fuel gas existing in the cooling medium tank **71** passes through the cooling tank internal gas discharge pipe **74** and is discharged from the gas discharge port **74a**. The fuel gas is light, and thus, the fuel gas discharged from the gas discharge port **74a** in the engine room **13** rises and is discharged outside the ship via the ventilation port **13a** in the engine room **13**.

(73) As the specified value, for example, 40% LEL may be adopted, but the specified value may be appropriately determined based on experiments or experience.

(74) On the other hand, when the cooling tank internal gas detector **73** detects that the concentration of the fuel gas in the cooling medium tank **71** is less than the specified value (No in **S1**), the control unit **12a** closes the cooling tank internal gas discharge valve **75** (**S3**). Thus, the inside of the cooling medium tank **71** is maintained in a sealed state.

(75) As described above, when the cooling tank internal gas detector **73** detects that the concentration of the fuel gas (a gaseous state of the fuel) in the cooling medium tank **71** is equal to or greater than the specified value determined in advance, the control unit **12a** opens the cooling tank internal gas discharge valve **75** (**S1**, **S2**). Thus, even if a fuel gas leak occurs in the fuel cell **31** for some reason and leaking fuel gas enters the cooling medium tank **71** via the cooling medium circulation pipe **72**, the fuel gas is discharged from the cooling tank internal gas discharge pipe **74** via the cooling tank internal gas discharge valve **75**. Therefore, it is possible to reduce the risk of leaking fuel gas staying in the cooling medium tank **71**.

(76) As described above, the gas discharge port **74a** of the cooling tank internal gas discharge pipe **74** is located above the electrical equipment EM in the engine room **13** in which the cooling medium tank **71** is installed (see FIG. 2). In this case, when the fuel gas leaking from the fuel cell **31** and entering the cooling medium tank **71** is discharged from the gas discharge port **74a** of the cooling tank internal gas discharge pipe **74**, the fuel gas having a low specific gravity (for example, hydrogen gas) rises, and thus, the fuel gas is unlikely to contact the electrical equipment EM located below. Thus, the risk of the discharged fuel gas being ignited by the electrical equipment EM can be reduced.

(77) The ventilation port **13a** of the engine room **13** is located above the gas discharge port **74a** (see FIG. 2). Thus, the fuel gas having a low specific gravity discharged from the gas discharge port **74a** rises from the gas discharge port **74a** as it is, and then is quickly (efficiently) discharged to the outside of the engine room **13** via the ventilation port **13a**. Therefore, it is possible to surely reduce the risk of the fuel gas being ignited by the electrical equipment EM in the engine room **13**.

(78) The fuel gas supply pipe **32** is a pipe through which the fuel gas passes, and thus, there is a possibility that the fuel gas leaks from the fuel gas supply pipe **32** for some reason. On the other hand, there is a possibility that the fuel gas leaking from the fuel cell **31** enters the cooling medium tank **71** via the cooling medium circulation pipe **72** and accumulates in the cooling medium tank **71**. Therefore, the fuel gas accumulated in the cooling medium tank **71** may also leak from the cooling medium tank **71** for some reason.

(79) As described in the present embodiment, the fuel gas supply pipe **32** is located to pass through the engine room **13** in which the cooling medium tank **71** is installed, so that components from which the fuel gas may leak (the fuel gas supply pipe **32** and the cooling medium tank **71**) are aggregated in the same space (the engine room **13**) in the hull **1**. Thus, a range having a risk for fuel gas leakage is smaller than in a case where the cooling medium tank **71** is installed in a place other than the engine room **13**, for example. Therefore, it is possible to increase the degree of freedom in design when installing electrical equipment outside the above-mentioned range. For example, even if an air supply fan is installed outside the ventilation port **13a** of the engine room **13** to actively ventilate the inside of the engine room **13**, the range in which the air supply fan can be installed is

large, and thus, the degree of freedom when installing the air supply fan can be increased.

(80) Incidentally, in the present embodiment, as illustrated in FIG. 2, the cooling medium tank 71 is connected to an uppermost part of the cooling medium circulation pipe 72. That is, the cooling medium tank 71 is located at an uppermost part of the circulation path of the cooling medium flowing through the cooling medium circulation pipe 72. When the positional relationship (connection relationship) between the cooling medium tank 71 and the cooling medium circulation pipe 72 are defined as described above, the following effects can be obtained.

(81) For example, in a case where the cooling medium tank 71 is connected to a part of the cooling medium circulation pipe 72 other than the uppermost part (for example, a lowermost part) of the cooling medium circulation pipe 72, when the control unit 12a opens the cooling tank internal gas discharge valve 75 according to the detection result of the fuel gas by the cooling tank internal gas detector 73, a “cooling medium leak” may occur in which the cooling medium in the cooling medium circulation pipe 72 flows into the cooling medium tank 71 by its own weight and flows out from the gas discharge port 74a of the cooling tank internal gas discharge pipe 74.

(82) When the cooling medium tank 71 is connected to the uppermost part of the cooling medium circulation pipe 72 to position the cooling medium tank 71 at the uppermost part of the circulation path of the cooling medium, it is possible to prevent the cooling medium in the cooling medium circulation pipe 72 from flowing into the cooling medium tank 71 by its own weight, even if the control unit 12a opens the cooling tank internal gas discharge valve 75. This makes it possible to prevent the cooling medium from leaking when the cooling tank internal gas discharge valve 75 is opened.

(83) When the cooling tank internal gas detector 73 detects that the concentration of the fuel gas in the cooling medium tank 71 is less than the specified value determined in advance, the control unit 12a closes the cooling tank internal gas discharge valve 75 to seal the cooling medium tank 71 (S3).

(84) When the operating temperature of the fuel cell 31 is high, it may be required to supply pressurized water (a pressurized cooling medium) to the fuel cell 31 to rapidly cool the fuel cell 31. For example, when the operating temperature of the fuel cell 31 exceeds 100° C., the liquid state (having a heat transfer coefficient higher by 10 times or more than that of the gaseous state) cannot be maintained unless the cooling medium is pressurized. If the fuel gas does not leak to the cooling medium tank 71, it is possible to easily supply the pressurized water, when the cooling medium tank 71 is sealed. That is, it is possible to pressurize the cooling medium in the cooling medium tank 71 and supply the pressurized cooling medium to the fuel cell 31. The cooling medium tank 71 is sealed, and thus, it is possible to avoid a problem that may occur when the upper part of the cooling medium tank 71 is always open. For example, it is also possible to prevent the cooling medium from flowing out to the outside of the tank due to shaking during navigation of the fuel cell ship SH, and to prevent impurities from mixing with the cooling medium inside the cooling medium tank 71.

(85) In particular, although the optimum operating temperature of the fuel cell 31 being a polymer electrolyte fuel cell is around 80° C., if the fuel cell 31 is continuously driven at a high load, the operating temperature of the fuel cell 31 may rise to 100° C. or higher. Therefore, the above-mentioned control, by which the cooling medium tank 71 is sealed to enable the supply of pressurized water when the fuel gas does not leak to the cooling medium tank 71, is very effective if the operating temperature of the fuel cell 31 is 100° C. or higher. That is, if at least one of the temperature of the cooling medium supplied to the fuel cell 31 and the temperature of the cooling medium discharged from the fuel cell 31 is 100° C. or higher, the above-mentioned control by which the cooling medium tank 71 is sealed to enable the supply of pressurized water is very effective.

(86) The temperature of the cooling medium supplied to the fuel cell 31 can be monitored by a supply-side cooling medium temperature detection unit 331a illustrated in FIGS. 5 and 6. On the

other hand, the temperature of the cooling medium discharged from the fuel cell **31** can be monitored by a discharge-side cooling medium temperature detection unit **331b**. The supply-side cooling medium temperature detection unit **331a** and the discharge-side cooling medium temperature detection unit **331b** are configured by a temperature sensor such as a thermistor.

(87) [4. Control to Prevent Fuel Gas Leakage in Fuel Cell]

(88) FIG. 5 is an explanatory diagram schematically illustrating a configuration of main parts of the fuel cell ship SH. The fuel cell ship SH may include a supply-side fuel gas pressure detection unit **332a** and a supply-side cooling medium pressure detection unit **333a**. The supply-side fuel gas pressure detection unit **332a** is a supply-side fuel pressure detection unit that detects a pressure $P1$ (MPa) of the fuel (for example, a fuel gas) supplied to the fuel cell **31** (from the fuel tank **41**). The supply-side cooling medium pressure detection unit **333a** detects a pressure $P2$ (MPa) of the cooling medium supplied to the fuel cell **31** (from the cooling medium tank **71**). The supply-side fuel gas pressure detection unit **332a** and the supply-side cooling medium pressure detection unit **333a** are each formed by a pressure sensor.

(89) When the pressure $P1$ of the fuel (for example, the fuel gas) is higher than the pressure $P2$ of the cooling medium according to the detection results of the supply-side fuel gas pressure detection unit **332a** and the supply-side cooling medium pressure detection unit **333a**, the control unit **12a** desirably stops the power generation by the fuel cell **31**.

(90) If $P1 > P2$, fuel may leak to the cooling medium side and mix with the cooling medium on the fuel gas supply side (an anode inlet side) in the fuel cell **31**. If $P1 > P2$, when the power generation by the fuel cell **31** is stopped, it is possible to suppress the fuel from leaking to the cooling medium side on the fuel supply side in the fuel cell **31**.

(91) FIG. 6 is an explanatory diagram schematically illustrating another configuration of main parts of the fuel cell ship SH. In addition to the supply-side fuel gas pressure detection unit **332a** and the supply-side cooling medium pressure detection unit **333a** illustrated in FIG. 5, the fuel cell ship SH may include a discharge-side fuel gas pressure detection unit **332b** and a discharge-side cooling medium pressure detection unit **333b**. The discharge-side fuel gas pressure detection unit **332b** is a discharge-side fuel pressure detection unit that detects a pressure $P3$ (MPa) of the fuel (for example, a fuel gas) discharged from the fuel cell **31**. The discharge-side cooling medium pressure detection unit **333b** detects a pressure $P4$ (MPa) of the cooling medium discharged from the fuel cell **31**. The discharge-side fuel gas pressure detection unit **332b** and the discharge-side cooling medium pressure detection unit **333b** are each formed by a pressure sensor.

(92) Here, the difference between the pressure $P1$ of the fuel (for example, the fuel gas) detected by the supply-side fuel gas pressure detection unit **332a** and the pressure $P2$ of the cooling medium detected by the supply-side cooling medium pressure detection unit **333a** is defined as a first pressure difference $Pd1$ (MPa). The difference between the pressure $P3$ of the fuel (for example, the fuel gas) detected by the discharge-side fuel gas pressure detection unit **332b** and the pressure $P4$ of the cooling medium detected by the discharge-side cooling medium pressure detection unit **333b** is defined as a second pressure difference $Pd2$ (MPa).

(93) The control unit **12a** desirably stops the power generation by the fuel cell **31**, when at least one of the first pressure difference $Pd1$ and the second pressure difference $Pd2$ is equal to or greater than a predetermined value Pth (MPa).

(94) When $Pd1 \geq Pth$, that is, when $(P1 - P2) \geq Pth$, for example, the supply pressure of the fuel is too much higher than the supply pressure of the cooling medium, so that the fuel may leak to the cooling medium side and mix with the cooling medium on the fuel supply side (the anode inlet side) in the fuel cell **31**. On the other hand, when $Pd2 \geq Pth$, that is, when $(P3 - P4) \geq Pth$, for example, the discharge pressure of the fuel is too much higher than the discharge pressure of the cooling medium, so that the fuel may leak to the cooling medium side and mix with the cooling medium on the fuel discharge side (the anode outlet side) in the fuel cell **31**.

(95) If the power generation by the fuel cell **31** is stopped when at least one of the conditions of

$Pd1 \geq P_{th}$ and $Pd2 \geq P_{th}$ is satisfied, it possible to suppress the risk of the fuel leaking to the cooling medium side on at least one of the fuel supply side and the fuel discharge side in the fuel cell **31**.
(96) In the present embodiment, a fuel gas being gaseous is used as the fuel supplied from the fuel tank **41** to the fuel cell **31**, but the fuel is not limited to a gas and may be a liquid. If a liquid fuel is used, when the liquid fuel leaks from a pipe, the leaked liquid fuel vaporizes and forms a gas (a fuel gas).

(97) Embodiments of the present invention have been described above, however, the scope of the present invention is not limited to these embodiments, and can be extended or modified without departing from the gist of the invention.

INDUSTRIAL APPLICABILITY

(98) The present invention can be utilized in a fuel cell ship, for example.

REFERENCE SIGNS LIST

(99) **1** . . . Hull **6** . . . Propulsion device **7** . . . Cooling system **12a** . . . Control unit **13** . . . Engine room (cooling medium tank installation compartment) **13a** . . . Ventilation port **31** . . . Fuel cell **32** . . . Fuel gas supply pipe (fuel supply pipe) **41** . . . Fuel tank **71** . . . Cooling medium tank **72** . . . Cooling medium circulation pipe **73** . . . Cooling tank internal gas detector **74** . . . Cooling tank internal gas discharge pipe **74a** . . . Gas discharge port **75** . . . Cooling tank internal gas discharge valve **332a** . . . Supply-side fuel gas pressure detection unit (supply-side fuel pressure detection unit) **333a** . . . Supply-side cooling medium pressure detection unit **332b** . . . Discharge-side fuel gas pressure detection unit (discharge-side fuel pressure detection unit) **333b** . . . Discharge-side cooling medium pressure detection unit EM . . . Electrical equipment SH . . . Fuel cell ship

Claims

1. A fuel cell ship comprising: a fuel cell configured to generate electric power by an electrochemical reaction of a fuel; a propulsion device configured to generate a propulsive force on a hull by the electric power supplied from the fuel cell; a cooling system configured to cool the fuel cell, the cooling system includes: a cooling medium tank that accommodates a cooling medium, a cooling medium circulation pipe configured to circulate the cooling medium between the fuel cell and the cooling medium tank, a cooling tank internal gas detector installed in the cooling medium tank, a cooling tank internal gas discharge pipe connected to the cooling medium tank, and a cooling tank internal gas discharge valve installed in the cooling tank internal gas discharge pipe; and a control unit configured to control opening and closing of the cooling tank internal gas discharge valve, wherein the control unit is further configured to open the cooling tank internal gas discharge valve based on the cooling tank internal gas detector detecting that a concentration of a gaseous fuel in the cooling medium tank is equal to or greater than a predetermined value.
2. The fuel cell ship according to claim 1, further comprising: a cooling medium tank installation compartment where the cooling medium tank is installed, wherein a gas discharge port of the cooling tank internal gas discharge pipe is located above electrical equipment in the cooling medium tank installation compartment.
3. The fuel cell ship according to claim 2, wherein: the cooling medium tank installation compartment includes a ventilation port configured to communicate with an outside of the fuel cell ship, and the ventilation port is located above the gas discharge port.
4. The fuel cell ship according to claim 2, further comprising: a fuel tank configured to accommodate the fuel; and a fuel supply pipe configured to supply the fuel from the fuel tank to the fuel cell, and wherein the fuel supply pipe passes through the cooling medium tank installation compartment.
5. The fuel cell ship according to claim 1, wherein the cooling medium tank is located at an uppermost part of a circulation path of the cooling medium flowing through the cooling medium circulation pipe.

6. The fuel cell ship according to claim 1, further comprising: a supply-side fuel pressure detection unit configured to detect a pressure of the fuel supplied to the fuel cell; and a supply-side cooling medium pressure detection unit configured to detect a pressure of the cooling medium supplied to the fuel cell, and wherein the control unit is configured to stop power generation by the fuel cell based on the pressure of the fuel being greater than the pressure of the cooling medium.
7. The fuel cell ship according to claim 1, further comprising: a supply-side fuel pressure detection unit configured to detect a pressure of the fuel supplied to the fuel cell; a supply-side cooling medium pressure detection unit configured to detect a pressure of the cooling medium supplied to the fuel cell; a discharge-side fuel pressure detection unit configured to detect a pressure of the fuel discharged from the fuel cell; and a discharge-side cooling medium pressure detection unit configured to detect a pressure of the cooling medium discharged from the fuel cell, wherein: a difference between the pressure of the fuel detected by the supply-side fuel pressure detection unit and the pressure of the cooling medium detected by the supply-side cooling medium pressure detection unit is defined as a first pressure difference, and a difference between the pressure of the fuel detected by the discharge-side fuel pressure detection unit and the pressure of the cooling medium detected by the discharge-side cooling medium pressure detection unit is defined as a second pressure difference, the control unit is further configured to stop power generation by the fuel cell based on the first pressure difference or the second pressure difference being equal to or greater than a second predetermined value.
8. The fuel cell ship according to claim 1, wherein the control unit is further configured to close the cooling tank internal gas discharge valve to seal the cooling medium tank based on the cooling tank internal gas detector detecting that the concentration of the gaseous fuel in the cooling medium tank is less than the predetermined value.
9. The fuel cell ship according to claim 8, wherein a temperature of the cooling medium supplied to the fuel cell and a temperature of the cooling medium discharged from the fuel cell is greater than or equal to 100° C.
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